Evaluation runs: **hermes** (hermes)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕒 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
3,2	7200	4652	conforms	2.1	4	4 h, 250 GB	4	7210	brain01, 2026-11-02
3,3	252865152		🕒 (4 h)			4 h, 250 GB			brain01, 2026-11-02

stateful

stateful: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
3,2	7200	4652	conforms	0.5	2	4 h, 250 GB	brain01, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 3 3 bag.

offline

offline: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	6	4652	3	violation	0.4	4	1 h, 250 GB	0.0	0.3	0.0	merge	brain01, 2026-11-03
3,3	252865152	90010608		conforms	2782.7	368	4 h, 250 GB	1860.6	922.0	0.0	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	6	4652	3	violation	28.6	2553	4 h, 250 GB	0.0	28.6	0.0	merge	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): 3 3 bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
3,2	6	4652	2310	violation	23.8	2647	4 h, 250 GB	136	brain01, 2026-11-03
3,3	90						4 h, 250 GB		brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	4652	4652	conforms	28.9	2553	4 h, 250 GB	28.6	0.3	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): 3 3 bag.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	4652	4652	conforms	28.9	2553	4 h, 250 GB	0.3	28.6	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 3 3 bag.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	4652	4652	conforms	24.0	2712	4 h, 250 GB		brain01, 2026-11-03
3,3	90010608		(4 h)			4 h, 250 GB		brain01, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	4652	4652	conforms	0.5	6	4 h, 250 GB		brain01, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: 3 3 bag.